

Final Comparison Table: Custom CNN vs MobileNetV2 vs ResNet50

Model	Accuracy	F1-Score	Training Time	Parameters	Grad-CAM Quality	SHAP Interpretability	Overall Finding
Custom CNN	74.4%	0.7438	~120s	19.4M	2/5	2/5 Good	Good accuracy but poor explainability; scattered attention & diffuse SHAP
MobileNetV2	74.6%	0.7453	~8min	2.6M	4/5	5/5	BEST: Highest accuracy, best explainability, patterns generalize well
ResNet50	66.4%	0.6648	~12min	24.1M	3/5	4/5 Lower	Lower accuracy; clean explanations but sometimes illogical feature importance